Azure HandsOn – Finance Container

To-Do´s

Prerequisites and expected deliveries

1. **Azure Subscription**
   You need to have an active Azure Student Subscription. You can get it by calling the following URL and follow the instructions: https://azure.microsoft.com/en-us/free/students/
   IMPORTANT: Use your FHNW account (e.g. Thomas.etteler@students.fhnw.ch)!

2. **Browser settings**
   Ensure that your Web Browser is displaying the Azure Portal in English.

3. **PeFiAs App**
   Deliver a running version of the personal vibe coded financial manager “PeFiAs” on the web, based on the tasks described in the section “Containers”.

General Information Sometimes you will see *product names* written in italic. This indicates that this represents the name of the resource you could look up in the Azure Marketplace to find it more quickly.

Containers

1. **Target Azure Region**
   Due to performance reasons and FHNW subscription limits you are kindly asked to create all Azure resources within the **Azure Region Sweden Central.**

2. **Resource Identifier**
   All resources may require a three-digit identifier, e.g. JSM for **J**ohn **Sm**ith. Please think of your personal identifier before you start with creating resources, which will save time later.

3. **Create your Resource Group**
   On your own Notebook, open a browser and call the Azure Portal at https://portal.azure.com. Login with your FHNW account that was used to claim the Azure Subscription. When you are logged in, create a new *resource group* with the name "RG-<JSM>-pefias" in the recommended Azure region, using e.g. the Azure search bar.

   Create your Linux based Container-Image creation VM
   Click "Create" in the upper left corner within the resource group. Type "virtual machine" in the portal search bar (at the top center), select the Virtual Machine offer from Microsoft and click "Create".
   Use the following details to create the VM

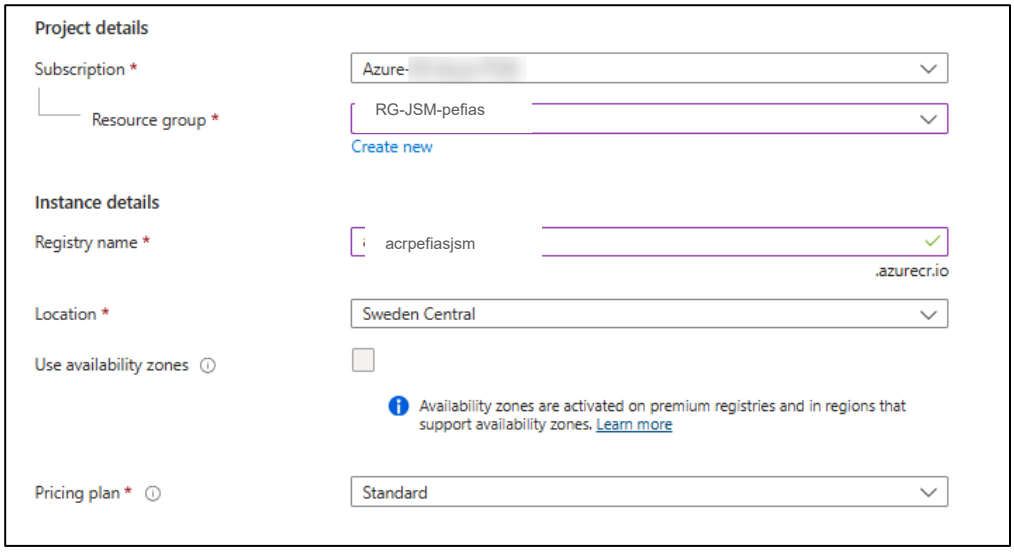
Name	vm-<JSM>-lin-cicreator
Availability options	No infra. red. Required
Security type	Standard
Image	Ubuntu Server 22.04 LTS
Size	D2s_v4 (**change it by clicking on "See all sizes" on the bottom**)
Authentication type	Password
Username	cloudhandson
Password	FHnw#1n-Cl4d (or your own **that you have to remember**!)

   All other values can stay in the default. You can click "Review+create" and once the validation finishes, click "Create".

   We do need this step in order to have a so called "Build Host" for the container image which leverages the application "Docker". Once this image is being built, we can push it to Azure and only then the Azure services can make use of it.

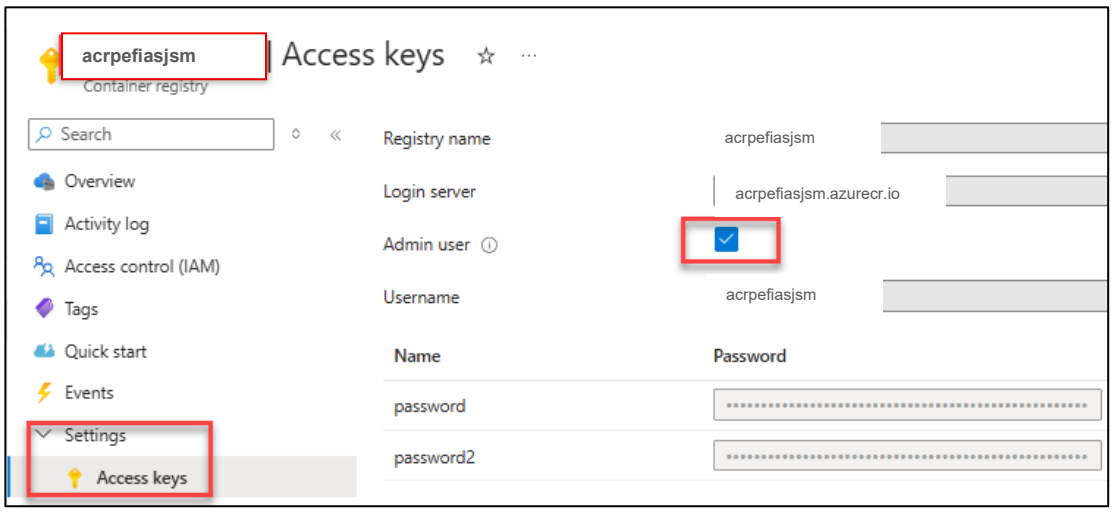
4. Create your Azure container registry (ACR)

Use the Azure portal search bar (top center) to create a new *container registry* that will reside in the resource group you have just created. Ensure to put it into the correct Azure Region.
Name: acrpefias<jsm> (be aware of the case sensitivity!)



All other settings can stay in the default.
Challenge if time permits: Try to find the corresponding Azure CLI Commands to achieve the same, but via Code.

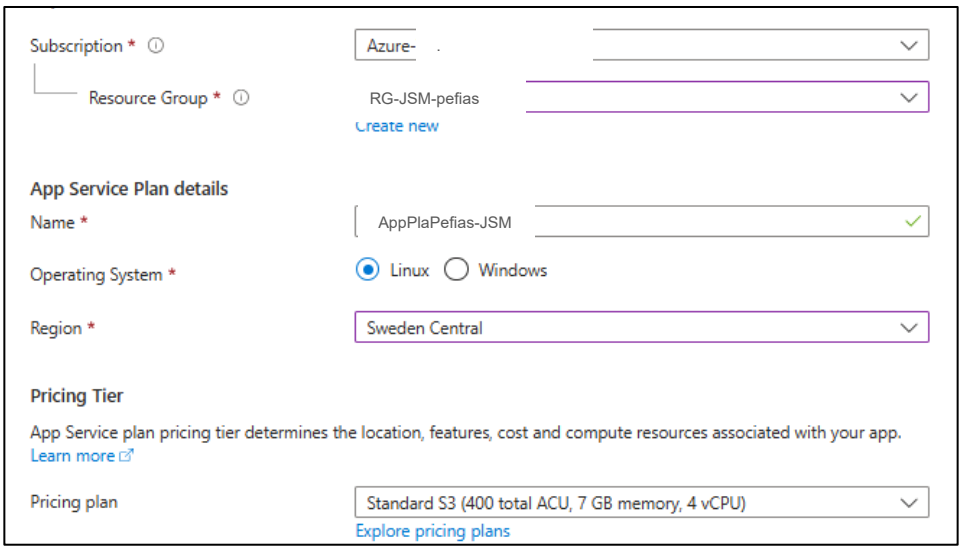
After the resource has been created successfully, ensure that the Admin user for the container registry is enabled:



5. Create your App Service Plan

Use the Azure search bar to create a new *app service plan* that will reside in the resource group you have created.

Resource Group:	RG-<JSM>-pefias
Name:	AppPlaPefias-<JSM>
Operating System:	Linux
Region:	recommended one from the beginning
Pricing Tier (SKU):	Standard S3 (find it through a click on “Explore pricing plans”)



The rest of the values can stay in standard and you can continue by clicking “Review+create” and finally click “Create”.

6. Assemble the container image and provide it in Azure

Connect to your Linux VM by using the public IP-Address of the Linux VM. You can find it in the Azure Portal at the VM Resource on the overview page in the upper right corner called “Public IP”.

The following commands apply to Windows and Mac users the same way.

Open the app “Terminal” and type

“`ssh cloudhandson@[public-IP-address]`” (exchange the IP Address!)

When you are asked for the credentials, use the following ones:

Username	cloudhandson
Password	FHnw#1n-Cl4d (or the one you have defined)

HINT (PLEASE READ!):
- Copy and Paste in “Terminal” works with a right click in the Window
- The copied password from the Clipboard WILL NOT appear in any shape or form after the “Password: “ sign in the “Terminal”. If you have right clicked ONCE while having the value of the password in your Notebooks Clipboard, just hint enter and you will be able to login to the VM.

Once you are in the Linux VM and have a prompt (command line), clone the GIT repository which contains the image to your local filesystem by using the following command:

```
git clone https://github.com/OliCSADoerr/PeFiAs
```

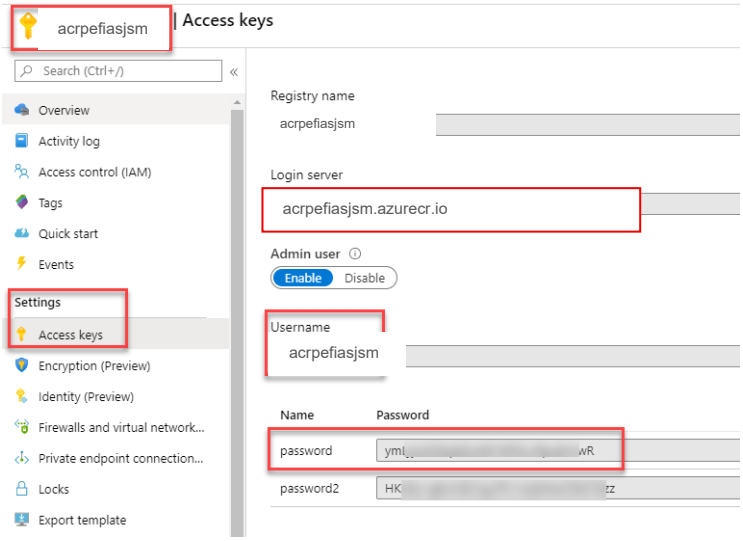
In order to build your container, you need to install docker on the Linux VM. This is done by using the following commands:

```
sudo apt-get update
sudo snap install docker
```

To build the container-image from the source code, you need to run the following commands. (ATTENTION – The name of the ACR is CaSe-SensiTiVe!!!):

```
cd PeFiAs/
sudo docker build -t "<your-acr-name>.azurecr.io/pefias:v1" -f Dockerfile .
```

To let Docker connect to your ACR, you need a username and a password. Therefore you will need the access credentials to the ACR. Go to your ACR in the Azure Portal and check the navigation pane on the left side. There is the section "Access Keys" from which you will get the required values as seen in the screenshot aside



Copy the values of the

- Login server
- Username
- Password (one of the two)

On your "Terminal", enter the following command to login into your ACR and provide the username and password you have just copied from the step above:
ATTENTION – The ACR name is CaSe-SensiTiVe and the invisible password pattern from the Linux VM login behaviour applies as well!!!

```
sudo docker login "<ACR-Name>.azurecr.io"
```

The image build will take a bit *– a good opportunity for having a break.*

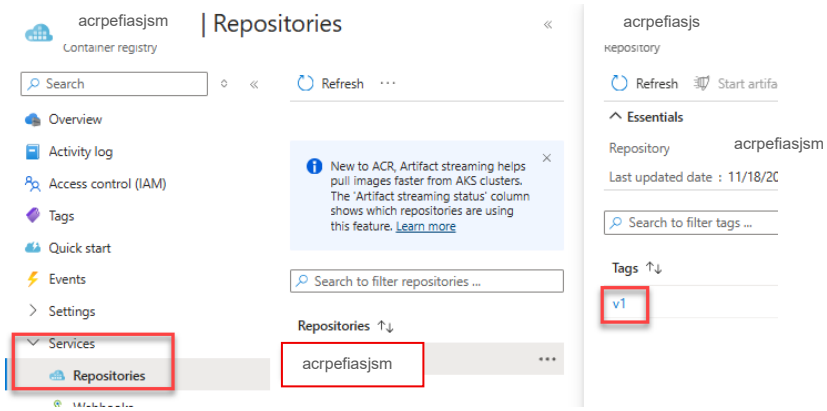
After that, only one small step is missing from the final goal – to have the image in the cloud. For that, enter the push command which moves the image into the cloud (your ACR resource):

```
sudo docker push "<your-acr-name>.azurecr.io/pefias:v1"
```

If you are more interested in the details behind the docker build command, then checkout the official documentation.

Validating the pushed container

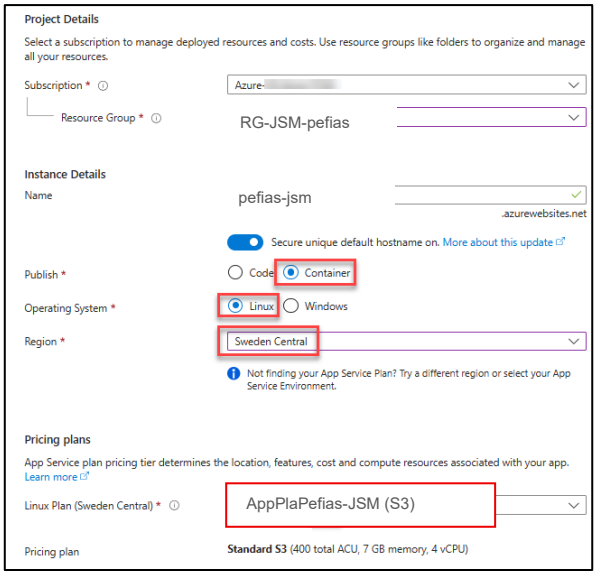
Once the push of the container completed, validate the image availability in your ACR. Therefore use the Azure Portal, navigate to the resource group from step 3, go to the ACR and check if you have an entry in the section “Repositories” within the left navigation pane of the ACR. It should look like this:

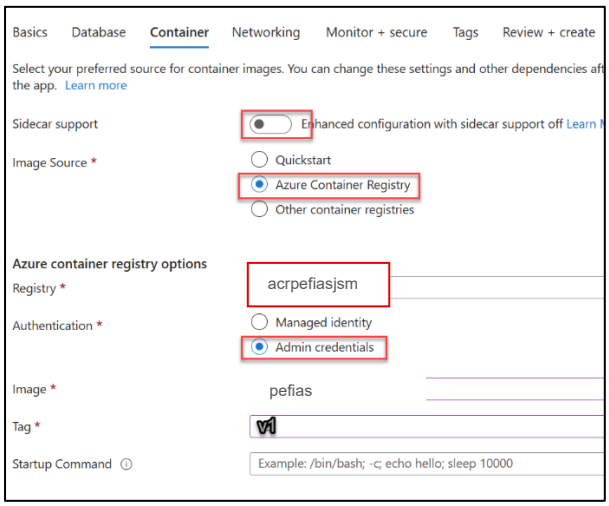


If that´s not the case, you have certainly missed some errors that popped up during the container creation. Therefore check the logs/output of the Azure CLI to find the error.

7. Create your Web App for Containers

Use the Azure search bar to create a new *App Service* that resides in the resource group you have created. In the Create Option select “WebApp”. Use the default settings, except the following values:

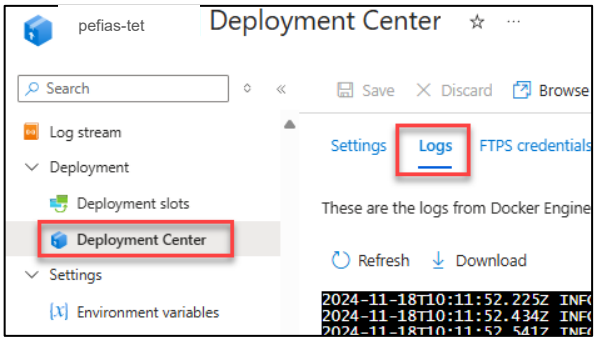




Name:	pefias-<JSM>	Sidecar support	Disabled
App Service Plan:	From step 7 (ensure it is S3)	Image Source:	**Azure Container Registry**
		Registry:	From step 4
		Authentication	Admin Credentials
		Image:	pefias
		Tag:	v1

Once those values are entered, go to “Review+create” and click the “Create” Button on the bottom left.

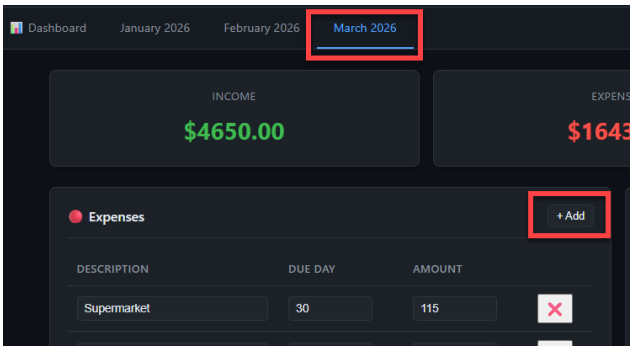
NOW – be patient! Usually it takes approximately up to 2 minutes until the new environment is up and running and the changes take effect.
IF – you are impatient - you can check the startup by consulting the container logs (Click on “Logs” in the section navigation) which you will find in the “Deployment Center” within the left navigation pane of the App Service.



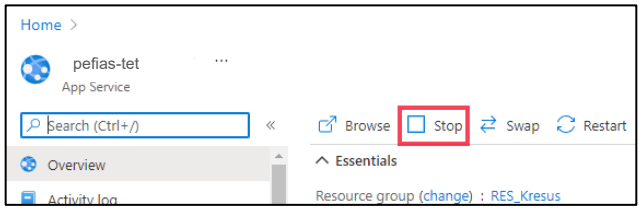
8. Validate your App and test it

Check if the Container is up and running and presents the starting dashboard “PeFiAs” by visiting the URL of the App Service you have just created.
You can find this URL on the overview page of the App Service resource in the upper right area – called “Default domain”.

Now add a new expense for March, by navigating to this month in the upper navigation bar and click “+ Add”.



Remember what you have inserted and stop the application using the controls in the Azure Portal for the App Service.
After that, start it again.
It is important that you do NOT use the Restart Feature!



Once you have validated that the server is up and running, using the WebApp logs in the Azure portal, open a new incognito browser window (to avoid cookie caching) and call the URL of your web application.

Check if the entered operation is still in the history.
If it is not there, think of a reason why this could be an expected outcome.
Are there any general concerns with that solution? If yes, suggest improvements.

9. Stop your resources – if this is the end

You could delete your resources easily by deleting the Resource group in which you have placed on all resources in.
This could be achieved easily by leveraging the following Azure CloudShell command:

```
az group delete --no-wait --yes --name RG-<JSM>-pefias
```